# Cayley-Hamilton theorem

## Basic skills

Q1. For M $|M-\lambda I|=0$ is the charateristic equation

Q2. a) $M=\begin{pmatrix}2&1\\0&2\end{pmatrix}$ $|M-\lambda I|=0$

subtract $\lambda$ on leading diag

$\Rightarrow \begin{vmatrix}2-\lambda & 1\\ 0 & 2-\lambda\end{vmatrix}=0 \Rightarrow (2-\lambda)^2=0$

b) $M=\begin{pmatrix}-1&3\\1&5\end{pmatrix}$ $|M-\lambda I|=0$

$\Rightarrow \begin{vmatrix}-1+\lambda & 3\\ 1 & 5-\lambda\end{vmatrix}=0 \Rightarrow (-1+\lambda)(5-\lambda)-3=0$

$\Rightarrow -\lambda^2+6\lambda-5-3=0 \Rightarrow \lambda^2-6\lambda+8=0$

c) $M=\begin{pmatrix}5&1&4\\0&4&-1\\2&6&-2\end{pmatrix}$ $|M-\lambda I|=0$

AS $\begin{vmatrix}5-\lambda & 1 & 4\\ 0 & 4-\lambda & -1\\ 2 & 6 & -2-\lambda\end{vmatrix} = (5-\lambda)\begin{vmatrix}4-\lambda & -1\\ 6 & -2-\lambda\end{vmatrix} - \begin{vmatrix}0 & -1\\ 2 & -2-\lambda\end{vmatrix}$

$+4\begin{vmatrix}0 & 4-\lambda\\ 2 & 6\end{vmatrix}$

$$\Rightarrow (5-\lambda)\left((4-\lambda)(-2-\lambda)+6\right)-(0+2)$$
$$+4\left(0-2(4-\lambda)\right)=0$$
$$\Rightarrow (5-\lambda)(\lambda^2-2\lambda-2)-2-8(4-\lambda)=0$$
$$\Rightarrow -\lambda^3+7\lambda^2-8\lambda-10-34+8\lambda=0$$
$$\Rightarrow -\lambda^3+7\lambda^2-44=0 \Rightarrow \lambda^3-7\lambda^2+44=0$$

Q3. $M^2-6M+3I_2=0$

a) $M^2=6M-3I_2$

b) $M^5=M^2M^3=6M^4-3M^3$

c) For $M^{-1}$ need $I_2$ so

$$I_2=\frac{6M-M^2}{3}=2M-\frac{1}{3}M^2$$

Then $M^{-1}=M^{-1}I_2=M^{-1}\left(2M-\frac{1}{3}M^2\right)$

$$=2I_2-\frac{1}{3}M$$

As $M^{-1}M=I_2$

## Competency skills

Q4. $M = \begin{pmatrix} 6 & 5 \\ -1 & 4 \end{pmatrix}$

a) $|M - \lambda I| = 0 \Rightarrow \begin{vmatrix} 6-\lambda & 5 \\ -1 & 4-\lambda \end{vmatrix} = 0$

so $(6-\lambda)(4-\lambda) + 5 = 0$

$\Rightarrow \quad \lambda^2 - 10\lambda + 29 = 0$

so by Cayley-Hamilton (replace $\lambda \to M$)

$M^2 - 10M + 29I_2 = 0$

b) $\times M^{-1}$ to get $M - 10I_2 + 29M^{-1} = 0$

so $M^{-1} = \frac{1}{29}(10I_2 - M)$

$= \frac{1}{29}\left(10\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 6 & 5 \\ -1 & 4 \end{pmatrix}\right)$

$= \frac{1}{29}\begin{pmatrix} 4 & -5 \\ 1 & 6 \end{pmatrix}$

Q5. $M = \begin{pmatrix} 0 & 2 \\ 1 & -2 \end{pmatrix} \quad |M - \lambda I| = 0$

$$\Rightarrow \quad -\lambda(-2-\lambda)-2=0 \Rightarrow \lambda^2+2\lambda-2=0$$

SO $\quad M^2+2M-2I_2=0$

SO $I_2 = M+\frac{1}{2}M^2$

$$\Rightarrow \quad M^{-1} = I_2 - \frac{1}{2}M = \begin{pmatrix}1&0\\0&1\end{pmatrix}+\frac{1}{2}\begin{pmatrix}0&2\\1&-2\end{pmatrix}$$

$$= \begin{pmatrix}1&1\\\frac{1}{2}&0\end{pmatrix} = \frac{1}{2}\begin{pmatrix}2&2\\1&0\end{pmatrix}$$

Q6. $\quad M = \begin{pmatrix}6&5&0\\0&2&-1\\0&1&-2\end{pmatrix} \quad |M-\lambda I|=0$

$$\begin{vmatrix}6-\lambda & 5 & 0\\ 0 & 2-\lambda & -1\\ 0 & 1 & -2-\lambda\end{vmatrix}=0$$

↓along column

$$\Rightarrow \quad (6-\lambda)\begin{vmatrix}2-\lambda & -1\\ 1 & -2-\lambda\end{vmatrix}=0$$

$$\Rightarrow \quad (6-\lambda)\left((2-\lambda)(-2-\lambda)+1\right)=0$$

$$\Rightarrow \quad (6-\lambda)(\lambda^2-3)=0 \Rightarrow -\lambda^3+6\lambda^2+3\lambda-18=0$$

SO $\quad \lambda^3-6\lambda^2-3\lambda+18=0$

Then by Cayley - Hamilton

$$M^3 - 6M^2 - 3M + 18I_3 = 0$$

a) so $M^3 = 6M^2 + 3M - 18I_3$

so need

$$M^2 = \begin{pmatrix} 6 & 5 & 0 \\ 0 & 2 & -1 \\ 0 & 1 & -2 \end{pmatrix}\begin{pmatrix} 6 & 5 & 0 \\ 0 & 2 & -1 \\ 0 & 1 & -2 \end{pmatrix} = \begin{pmatrix} 36 & 40 & -5 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

so $M^3 = 6\begin{pmatrix} 36 & 40 & -5 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} + 3\begin{pmatrix} 6 & 5 & 0 \\ 0 & 2 & -1 \\ 0 & 1 & -2 \end{pmatrix} - 18\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$$= \begin{pmatrix} 216 & 255 & -30 \\ 0 & 6 & -3 \\ 0 & 3 & -6 \end{pmatrix}$$

b) As $M^3 - 6M^2 - 3M + 18I_3 = 0$

$$\Rightarrow I_3 = \frac{1}{18}(6M^2 + 3M - M^3)$$

so $M^{-1} = \frac{1}{18}(6M + 3I_3 - M^2)$

$$\Rightarrow M^{-1} = \frac{1}{18}\left( 6\begin{pmatrix} 6 & 5 & 0 \\ 0 & 2 & -1 \\ 0 & 1 & -2 \end{pmatrix} + 3\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 36 & 40 & -5 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}\right)$$

$$= \frac{1}{18}\begin{pmatrix} 3 & -10 & 5 \\ 0 & 12 & -6 \\ 0 & 6 & -12 \end{pmatrix}$$

## Advanced skills

Q7. $M = \begin{pmatrix} 7 & -1 \\ 8 & 2 \end{pmatrix}$

a) $|M - \lambda I| = 0 \Rightarrow (7-\lambda)(2-\lambda) + 8 = 0$

$\Rightarrow \lambda^2 - 9\lambda + 22 = 0$ so $M^2 - 9M + 22I_2 = 0$

b) so $M^2 = 9M - 22I_2$

Then $M^n = M^2 M^{n-2} = 9MM^{n-2} - 22I_2 M^{n-2}$

$= 9M^{n-1} - 22M^{n-2}$

Q8. $M$ has $\lambda = -2, 3$ eigenvalues

a) so $|M + 2I| = 0$, $|M - 3I| = 0$

and $(\lambda - -2)(\lambda - 3) = 0 \Rightarrow \lambda^2 - \lambda - 6 = 0$

So $M^2 - M - 6I_2 = 0$

So $M^2 - M = 6I_2$

b) Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow M^2 = \begin{pmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{pmatrix}$

Then $\begin{pmatrix} a^2+bc-a & ab+bd-b \\ ac+cd-c & bc+d^2-d \end{pmatrix} = \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix}$

So
$$a^2 - a + bc = 6$$
$$d^2 - d + bc = 6$$
$$ab + bd - b = 0$$
$$ac + cd - c = 0$$

$\Rightarrow \quad a^2 - a = d^2 - d$

note. $a = d$ works but not only solution

also $b(a+d-1) = 0$, $c(a+d-1) = 0$

So $a = 1 - d$ or $b = c = 0$

and if $a = 1-d$ then $(1-d)^2 - (1-d) = d^2 - d$

$\Rightarrow d^2 - 2d + 1 - 1 + d = d^2 - d$

true for all $d$.

SO Not unique

$a=-2$, $d=3$
$b=c=0$
$\begin{pmatrix} -2 & 0 \\ 0 & 3 \end{pmatrix}$

or $a=3$, $d=-2$
$b=c=1$
$\begin{pmatrix} 3 & 1 \\ 1 & -2 \end{pmatrix}$

Q9. using $M=\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

Then $|M-\lambda I|=0 \Rightarrow (\lambda-a)(\lambda-d)-bc=0$

SO $\lambda^2-(a+d)\lambda+ad-bc=0$

SO show $M^2-(a+d)M+(ad-bc)I_2=0$

$M^2=\begin{pmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{pmatrix}$

SO

$$\begin{pmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{pmatrix}-(a+d)\begin{pmatrix} a & b \\ c & d \end{pmatrix}+\begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix}$$

$$= \begin{pmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{pmatrix} - \begin{pmatrix} a^2+ad & ba+bd \\ ac+dc & ad+d^2 \end{pmatrix}$$

$$+ \begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix}$$

$$= \begin{pmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{pmatrix} - \begin{pmatrix} a^2+bc & ba+bd \\ ac+dc & d^2+bc \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \text{ AS required.}$$